Kuboid va kubning bir yog'i yuzini topish

Kuboidning hajmi va bitta qirrasi ma'lum bo'lsa, uning bir yog'i yuzini o'lchamasdan topsa bo'ladi. Shu usul bilan idishdagi suv haqidagi masalalar ham yechiladi.

1-QISM

Mavzu

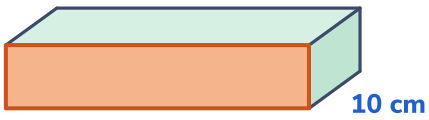
ESLAB OLAYLIK

- Kuboid hajmi = bo'yi × eni × balandligi = asosi yuzi × balandligi.
- Kvadratning tomoni = $\sqrt{\text{yuzi}}$, masalan $\sqrt{121} = 11$. Kubning qirrasi = $\sqrt[3]{\text{hajmi}}$, masalan $\sqrt[3]{1000} = 10$.
- $1\ \text{l} = 1000$ cm³.

Kuboidning bo'yi 24 cm, eni 10 cm, balandligi 5 cm. Hajmi: $24 \cdot 10 \cdot 5 = 1200$ cm³. Uning uchta yog'ini navbat bilan bo'yaymiz. Har gal bo'yalgan yoqning tomoni bo'lmagan qirrasini yozib qo'yamiz.



A



B



C

	Bo'yalgan yoq	Yozilgan qirra	Hajmi : qirrasi	Tekshirish
A	yon yoq	bo'yi 24 cm	$1200 : 24 = 50$ cm²	$10 \cdot 5 = 50$
B	old yoq	eni 10 cm	$1200 : 10 = 120$ cm²	$24 \cdot 5 = 120$
C	ust yoq	balandligi 5 cm	$1200 : 5 = 240$ cm²	$24 \cdot 10 = 240$

Hajm — yoq yuzi bilan uchinchi qirraning ko'paytmasi. Shuning uchun hajmni shu qirraga bo'lsak, yoqning yuzi chiqadi.

USUL · KUBOIDNING HAJMI VA QIRRASIGA KO'RA YOG'INING YUZINI TOPISH

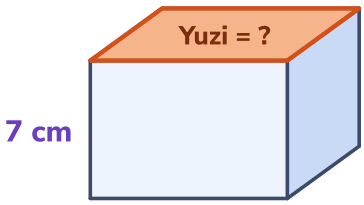
$$\text{Kuboid yog'ining yuzi} = \frac{\text{hajmi}}{\text{qirrasi}}$$



Qaysi qirraga bo'lamiz? Bo'yalgan yoqning to'rt tomoniga kirmagan qirraga.

1-MISOL · UST YOG'I

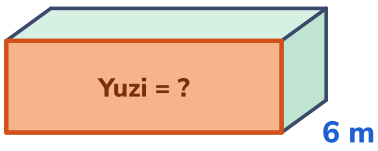
Kuboidning hajmi 630 cm³, balandligi 7 cm. Bo'yalgan yog'ining yuzini toping.



1. Ust yoq bo'yi va enidan iborat. Unga kirmagan qirra — balandligi 7 cm.
2. Yuzi $= \frac{630}{7} = 90$ cm².

2-MISOL · OLD YOG'I

Kuboidning hajmi 450 m³, eni 6 m. Bo'yalgan yog'ining yuzini toping.



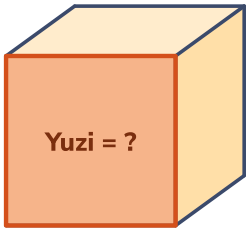
1. Old yoq bo'yi va balandligidan iborat. Unga kirmagan qirra — eni 6 m.
2. Yuzi $= \frac{450}{6} = 75$ m².

USUL · KUBNING HAJMIGA KO'RA YOG'INING YUZINI TOPISH

$$\text{Qirra} = \sqrt[3]{\text{hajmi}} \quad \text{Kub yog'ining yuzi = qirrasi × qirrasi}$$

3-MISOL · KUB

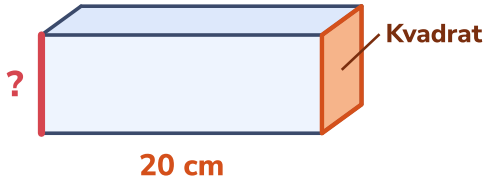
Kubning hajmi 2197 cm³. Uning bir yog'i yuzini toping.



1. Qirrasi $= \sqrt[3]{2197} = 13$ cm (kalkulyatorda kub ildiz tugmasi bilan topiladi). Tekshiramiz: $13 \cdot 13 \cdot 13 = 2197$.
2. Yog'ining yuzi $= 13 \cdot 13 = 169$ cm².

4-MISOL · KVADRAT SHAKLIDAGI YOQ

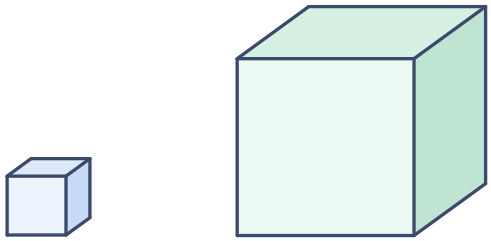
Kuboidning hajmi 980 cm³, bo'yi 20 cm. Uning ikkita qarama-qarshi yon yog'i kvadrat. Kvadratning tomonini toping.



1. Kvadrat yoqning yuzi $= \dfrac{980}{20} = 49$ cm².
2. Kvadratning tomoni $= \sqrt{49} = 7$ cm.

5-MISOL · QIRRASI BIR NECHA MARTA UZUN KUB

A kubning hajmi 1000 cm³. B kubning qirrasi A kubning qirrasidan 3 marta uzun. B kubning hajmini toping.

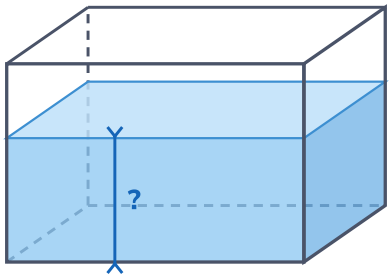


A B

1. A kubning qirrasi $= \sqrt[3]{1000} = 10$ cm.
2. B kubning qirrasi $= 3 \cdot 10 = 30$ cm.
3. B kubning hajmi $= 30 \cdot 30 \cdot 30 = 27\,000$ cm³.

6-MISOL · SUV SATHINING BALANDLIGI

Akvarium asosining yuzi 2400 cm². Unga 60 l suv quyildi. Suv sathining balandligi qancha?



1. $60\ \text{l} = 60 \cdot 1000 = 60\,000$ cm³.
2. Suv sathining balandligi $= \dfrac{60000}{2400} = 25$ cm.

USUL · KUBOID SHAKLIDAGI IDISHDAGI SUV

$$\text{Suv sathining balandligi} = \frac{\text{hajm}}{\text{asosining yuzi}}$$

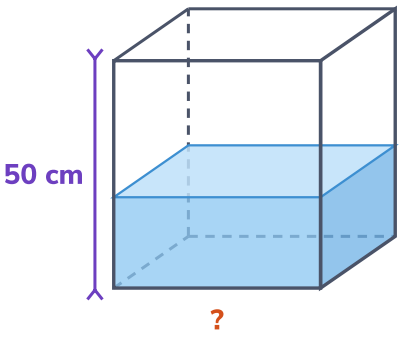
$$\text{Idish asosining yuzi} = \frac{\text{hajmi}}{\text{balandligi}}$$

Idishdagi suv ham kuboid shaklida. Uning asosi — idish tubi, balandligi — suv sathi.



7-MISOL · IDISHNING BIR QISMI TO'LGAN

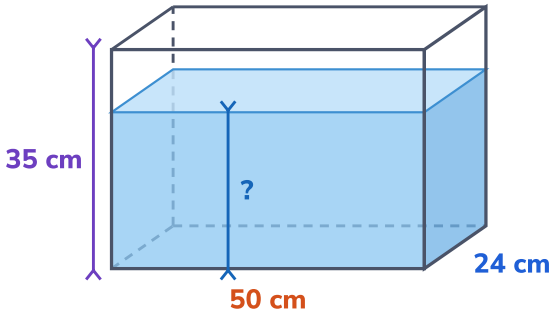
Idishning balandligi 50 cm, asosi kvadrat. Idishning $\frac{2}{5}$ qismi to'lishi uchun 32 l suv ketdi. a) Idish asosining yuzini toping. b) Asosining tomonini toping.



1. Suv sathi $= \frac{2}{5} \cdot 50 = 20$ cm. $32\ \text{l} = 32\,000$ cm³.
2. a) Asosining yuzi $= \frac{32000}{20} = 1600$ cm².
3. b) Asosining tomoni $= \sqrt{1600} = 40$ cm.

8-MISOL · TO'LISHI UCHUN YANA QANCHA SUV KERAK

Idishning bo'yi 50 cm, eni 24 cm, balandligi 35 cm. Unga 30 l suv quyildi. a) Suv sathi qanday balandlikda? b) Idish to'lishi uchun yana necha litr suv kerak?



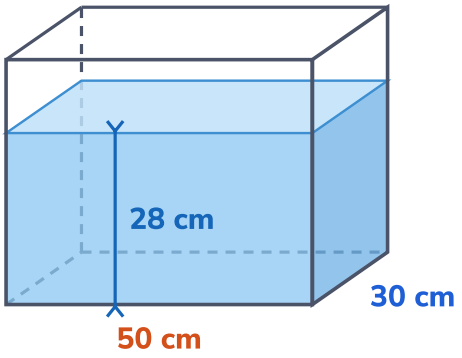
1. Asosining yuzi $= 50 \cdot 24 = 1200$ cm². $30\ \text{l} = 30\,000$ cm³.
2. a) Suv sathi $= \frac{30000}{1200} = 25$ cm.
3. b) Bo'sh qism balandligi $35 - 25 = 10$ cm. Uning hajmi $1200 \cdot 10 = 12\,000$ cm³ $= 12$ l.

2-QISM

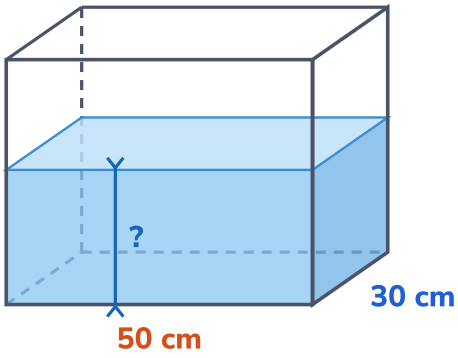
Misollar ishlash

1-MISOL · SUVNING BIR QISMI OLINDI

Akvariumning tubi 50 cm ga 30 cm. Undagi suv sathi 28 cm. Akvariumdan 9 l suv olindi. Endi suv sathi qanday balandlikda?



oldin



9 l olingandan keyin

1-usul. Avval sath qanchaga tushganini topamiz.

1. Tubining yuzi $= 50 \cdot 30 = 1500$ cm². $9\ \text{l} = 9000$ cm³.
2. Sath $\dfrac{9000}{1500} = 6$ cm ga tushdi. Endi sath $28 - 6 = 22$ cm.

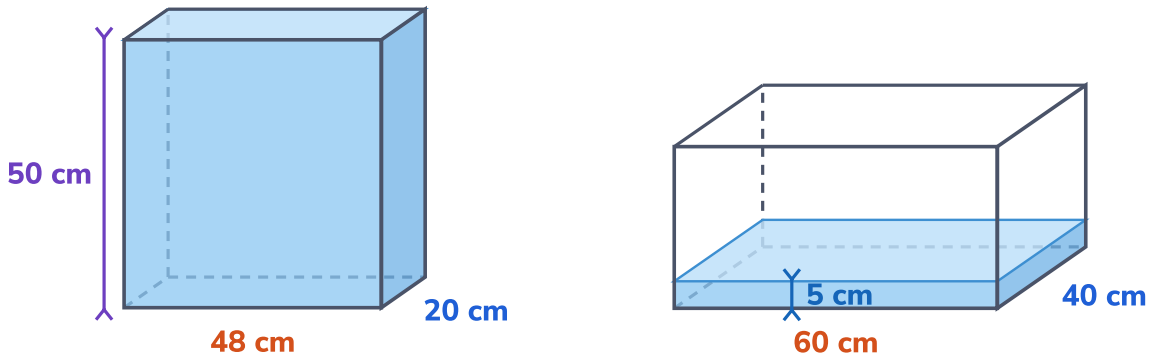
2-usul. Avval qolgan suvning hajmini topamiz.

1. Oldin suv: $1500 \cdot 28 = 42\,000$ cm³. Qoldi: $42\,000 - 9000 = 33\,000$ cm³.
2. Endi sath $= \dfrac{33000}{1500} = 22$ cm.

Suv qo'shilsa ham shunday: sath necha cm ga ko'tarilganini topib, oldingi sathga qo'shamiz.

2-MISOL · SUV BOSHQA IDISHGA QUYILDI

A idishning bo'yi 48 cm, eni 20 cm, balandligi 50 cm, u suvga to'la. B idishning tubi 60 cm ga 40 cm, unda 5 cm balandlikda suv bor. A idishdagi hamma suv B idishga quyildi. B idishdagi suv sathi qancha bo'ldi?



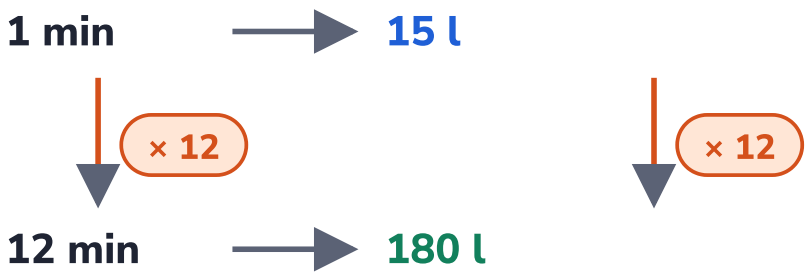
A idish B idish

1. A dagi suv: $48 \cdot 20 \cdot 50 = 48\,000$ cm³. Quyilganda suv hajmi o'zgarmaydi.
2. B tubining yuzi $= 60 \cdot 40 = 2400$ cm². Sath $\dfrac{48000}{2400} = 20$ cm ga ko'tariladi.
3. B dagi sath $= 5 + 20 = 25$ cm.

Agar A idishning faqat bir qismi to'la bo'lsa, avval o'sha qismdagi suv hajmini topamiz.

3-MISOL · JO'MRAKDAN SUV OQADI

Bo'sh idishning tubi 90 cm ga 40 cm. Jo'mrakdan unga har minutda 15 l suv tushadi. a) Idish 12 minutda to'ldi. Idishning balandligi qancha? b) Suv sathi 25 cm bo'lishi uchun necha minut kerak?



1. a) 12 minutda: $15 \cdot 12 = 180$ l $= 180\,000$ cm³. Tubining yuzi $= 90 \cdot 40 = 3600$ cm².

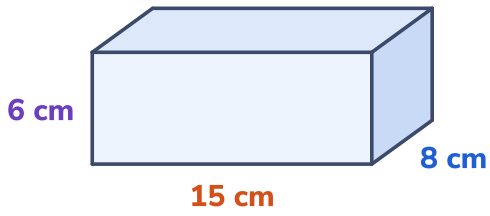
2 Balandligi $= \dfrac{180000}{3600} = 50$ cm.

3 b) 25 cm gacha suv: $3600 \cdot 25 = 90\,000$ cm³ $= 90$ l. Vaqt: $90 : 15 = 6$ minut.

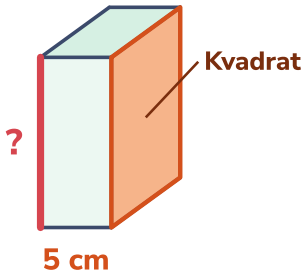
Suv idishdan oqib chiqsa ham shunday hisoblanadi.

4-MISOL · HAJMI TENG KUBOIDLAR

P kuboidning bo'yi 15 cm, eni 8 cm, balandligi 6 cm. Q kuboidning hajmi P kuboidnikiga teng, bo'yi 5 cm. Q ning ikkita qarama-qarshi yon yog'i kvadrat. a) Kvadrat yoqning yuzini toping. b) Kvadratning tomonini toping.



P



Q

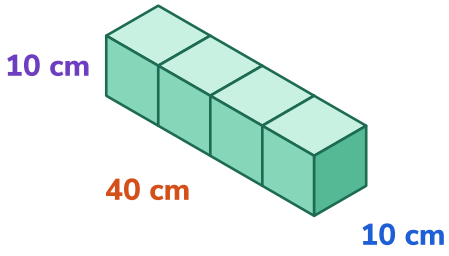
1 P ning hajmi $= 15 \cdot 8 \cdot 6 = 720$ cm³. Q ning hajmi ham 720 cm³.

2 a) Kvadrat yoqning yuzi $= \dfrac{720}{5} = 144$ cm².

3 b) Tomoni $= \sqrt{144} = 12$ cm.

5-MISOL · KUBLARDAN YASALGAN JISM

Har birining hajmi 1000 cm³ bo'lgan 4 ta kub bir qatorga yelimlandi. a) Hosil bo'lgan kuboidning o'lchamlari va hajmini toping. b) Kuboidning hamma tomoni bo'yaldi. Bo'yalgan yuz qancha?



1 Bitta kubning qirrasi $= \sqrt[3]{1000} = 10$ cm, yog'ining yuzi $= 10 \cdot 10 = 100$ cm².

2 a) Kuboid: bo'yi $4 \cdot 10 = 40$ cm, eni 10 cm, balandligi 10 cm. Hajmi $= 4 \cdot 1000 = 4000$ cm³.

3 b) 4 ta kubda $4 \cdot 6 = 24$ ta yoq bor. Ikki qo'shni kub tegib turgan joyda ikkala kubning bittadan yog'i ichkarida qoladi. Bunday joy 3 ta: $3 \cdot 2 = 6$ ta yoq ko'rinmaydi.

4 Tashqarida $24 - 6 = 18$ ta yoq. Bo'yalgan yuz $= 18 \cdot 100 = 1800$ cm².

6-MISOL · QUTIGA NECHTA BRUSOK SIG'ADI

Qutining bo'yi 30 cm, eni 20 cm, balandligi 12 cm. a) Unga bo'yi 10 cm, eni 5 cm, balandligi 4 cm bo'lgan brusoklar bir xil holatda teriladi. Qutiga nechta brusok sig'adi? b) Qirrasi 8 cm bo'lgan kubchalardan nechtasi sig'adi?

1 a) Har o'lcham bo'yicha: $30 : 10 = 3$, $20 : 5 = 4$, $12 : 4 = 3$. Brusoklar soni $= 3 \cdot 4 \cdot 3 = 36$.

2 b) $30 : 8 = 3$ (qoldiq 6), $20 : 8 = 2$ (qoldiq 4), $12 : 8 = 1$ (qoldiq 4). Qoldiqqa butun kubcha sig'maydi. Kubchalar soni $= 3 \cdot 2 \cdot 1 = 6$.

b) da hajmlarni bo'lish noto'g'ri javob beradi: $7200 : 512$ — 14 dan ko'p chiqadi, lekin bo'sh joylarga butun kubcha sig'maydi.

Kubning qirrasi 3 marta uzaysa, hajmi 3 marta emas, $3 \cdot 3 \cdot 3 = 27$ marta kattalashadi.



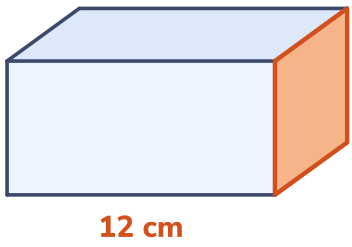
DIQQAT!

1. Hajmni bo'yalgan yoqqa kirmagan qirraga bo'ling. Ust yoq uchun: ✗ $1200 : 24 = 50$ ✓ $1200 : 5 = 240$ cm²
2. Kub yog'ining yuzini topishda hajmni 3 ga yoki 6 ga bo'lmang: ✗ $2197 : 3$ ✓ $\sqrt[3]{2197} = 13,\ \ 13 \cdot 13 = 169$ cm²
3. Kvadratning tomoni — yuzining ildizi, yarmi emas: ✗ $49 : 2$ ✓ $\sqrt{49} = 7$ cm
4. Litrni cm³ ga aylantiring: ✗ $60 : 2400$ ✓ $60\,000 : 2400 = 25$ cm
5. Suv olinganda yoki qo'shilganda oldingi sathni unutmang: ✗ sath 6 cm ✓ $28 - 6 = 22$ cm

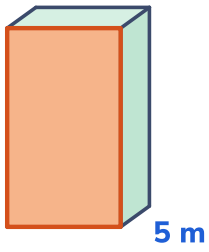
3-QISM

Mustahkamlash

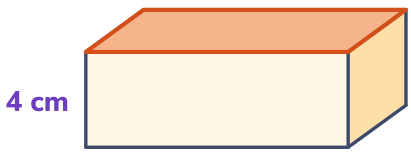
1. Kuboidning bo'yalgan yog'i yuzini toping.



a) Hajmi 576 cm³



b) Hajmi 560 m³



c) Hajmi 264 cm³

2. Kubning hajmi berilgan. Uning bir yog'i yuzini toping. Kub ildizni kalkulyatorda topsa bo'ladi.
   a) 27 000 cm³; b) 9261 cm³; c) 12 167 m³.
3. Kuboid shaklidagi stol tortmasining hajmi 36 000 cm³, chuqurligi (eni) 40 cm. Tortma old yog'ining yuzini toping.

4. Jadvaldagi bo'sh kataklarni to'ldiring.

Kuboid	Hajmi (cm^3)	Bo'yalgan yoq yuzi (cm^2)	Yoqqa kirmagan qirra (cm)
A	420	?	12
B	960	40	?
C	231	21	?
D	600	?	15

5. Asosi kvadrat bo'lgan ustunning hajmi 1125 cm^3, balandligi 45 cm.
   a) Asosining yuzini toping. b) Asosidagi kvadratning tomonini toping.

6. Kuboidning hajmi 1350 cm^3, bo'yi 6 cm. Uning ikkita qarama-qarshi yon yog'i kvadrat. Kvadratning tomonini toping.

7. Kub shaklidagi qutichaning hajmi 125 cm^3. Kub shaklidagi katta qutining qirrasi qutichanikidan 2 marta uzun.
   a) Katta qutining hajmini toping. b) Katta qutining hajmi qutichanikidan necha marta katta?

8. Hajmi 864 cm^3 bo'lgan yog'och brus 4 ta bir xil kubga arralandi.
   a) Bitta kubning qirrasini toping. b) Brusning bo'yi qancha edi?

9. Hovuz tubining yuzi 18 m^2. Unga 27 m^3 suv quyildi. Suv qanday chuqurlikda?

10. Kuboid shaklidagi kanistrga to'la quyilganda 24 l suv sig'adi, tubining yuzi 800 cm^2.
    a) Kanistrning balandligi qancha? b) Kanistrga 18 l suv quyilsa, suv sathi qanday balandlikda bo'ladi?

11. Bakning asosi — tomoni 30 cm bo'lgan kvadrat. Bakda 18 l suv bor. Bak to'lishi uchun yana 27 l suv kerak. Bakning balandligini toping.

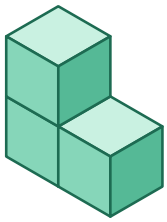
12. Idishning bo'yi 48 cm, eni 25 cm, balandligi 32 cm. Unga uch chelakdan 6 l dan suv quyildi.
    a) Suv sathi qanday balandlikda? b) Idish to'lishi uchun yana necha litr suv kerak?

13. Hovuzchaning tubi 2 m ga 1,5 m, undagi suv sathi 0,9 m. Nasos hovuzchadan 1,2 m^3 suv tortib oldi. Endi suv sathi qanday balandlikda?

14. Bo'yi 30 cm, eni 20 cm, balandligi 20 cm bo'lgan idishning yarmigacha olma sharbati bor. Hamma sharbat tubi 50 cm ga 20 cm bo'lgan bo'sh idishga o'tkazildi. Ikkinchi idishda sharbat qanday balandlikda turadi?

15. Bo'sh idishning tubi 60 cm ga 40 cm, balandligi 50 cm. Unga har minutda 8 l suv quyilmoqda.
    a) 9 minutdan keyin suv sathi qanday balandlikda bo'ladi? b) Idish necha minutda to'ladi?

16. Har birining hajmi 27 cm^3 bo'lgan 3 ta kubdan rasmdagi jism yasaldi. Jismning hamma tomoni (pasti ham) bo'yaldi. Bo'yalgan yuzni toping.



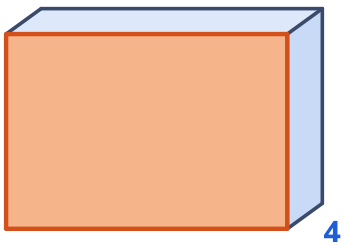
17. Gul ekiladigan yashik kuboid shaklida: bo'yi 50 cm, ikkita qarama-qarshi yon yog'i kvadrat. Yashikni yarmigacha to'ldirishga 10 l tuproq ketdi. Yashikning enini toping.

18. Katta idish tubining yuzi 900 cm^2, undagi suv sathi 32 cm. Shu suv tubining yuzi 600 cm^2 bo'lgan bo'sh idishga quyilganda, u to'la to'ldi.
    a) Ikkinchi idishning balandligi qancha? b) Katta idishning balandligi 40 cm. Suv quyib olinishidan oldin katta idish to'lishi uchun yana necha litr suv kerak edi?

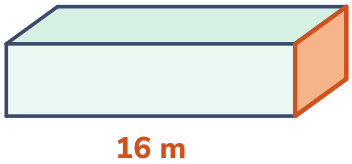
4-QISM

Mustaqil ish

1. Kuboidning bo'yalgan yog'i yuzini toping.



a) Hajmi 468 cm^3



b) Hajmi 448 m^3

2. Kubning hajmi 10 648 cm^3. Uning bir yog'i yuzini toping. (Kub ildizni kalkulyatorda toping.)

3. Asosi kvadrat bo'lgan kuboidning hajmi 1620 cm³, balandligi 5 cm. Asosidagi kvadratning tomonini toping.

4. Tosh plitaning bo'yi 30 cm, eni 20 cm, qalinligi 6 cm. Shu hajmdagi ustunning bo'yi 36 cm, ikkita qarama-qarshi yon yog'i kvadrat. Kvadratning tomonini toping.

5. Katta kubning hajmi 13 824 cm³. Uning qirrasi kichik kubning qirrasidan 4 marta uzun. Kichik kubning hajmini toping. (Kub ildizni kalkulyatorda toping.)

6. Akvariumning bo'yi 70 cm, eni 40 cm, balandligi 45 cm. Uni to'ldirish uchun yana 42 l suv kerak.
   a) Akvariumdagi suv sathi qanday balandlikda? b) Akvariumda necha litr suv bor?

7. Yomg'ir suvi yig'iladigan bakning tubi 40 cm ga 30 cm. Kechqurun bakdagi suv sathi 12 cm edi. Tunda yomg'irdan bakka 6 l suv tushdi. Ertalab suv sathi qancha bo'ldi?

8. Fermada bo'yi 50 cm, eni 40 cm, balandligi 30 cm bo'lgan idishning $\frac{2}{3}$ qismi sutga to'la. Hamma sut tubi 50 cm ga 50 cm bo'lgan bo'sh tog'oraga quyildi. Tog'orada sut qanday balandlikda turadi?

9. Suvga to'la idishning tubi 80 cm ga 45 cm, balandligi 40 cm. Jo'mrak ochildi, har minutda 9 l suv oqib chiqadi.
   a) Idish necha minutda bo'shaydi? b) 4 minutdan keyin suv sathi qanday balandlikda bo'ladi?

10. Qutining bo'yi 40 cm, eni 30 cm, balandligi 20 cm. Unga bo'yi 10 cm, eni 6 cm, balandligi 5 cm bo'lgan sovun qutichalari bir xil holatda teriladi. Nechta quticha sig'adi?

11. Javonga kub shaklidagi quti (hajmi 216 cm³) va bo'yi 10 cm bo'lgan uzun quti bir-biriga zich taqab qo'yildi. Uzun qutining yon yog'i kub yog'i bilan aynan bir xil kvadrat, shuning uchun ikkalasi birga bitta kuboid shaklida bo'ldi. Shu kuboidning bo'yi va hajmini toping.

12. **Xatoni toping.** Kubning hajmi 1000 cm³.
    a) Nodir bir yog'ining yuzini $1000 : 6$ deb hisobladi. Xato nimada? To'g'ri javob qancha?
    b) Sevara qirrani topish uchun kalkulyatorda $\sqrt{1000}$ ni hisobladi va 31,6 chiqdi. Xato nimada?

13. **Qiyinroq.** Do'konda kuboid shaklidagi ikki bo'lak pishloq bor. Ikkalasining eni 10 cm, balandligi bir xil. Birinchisining bo'yi 12 cm, ikkinchisiniki 20 cm. Ikkinchi bo'lakning hajmi birinchisinikidan 640 cm³ ko'p. Bo'laklarning balandligini toping. (Hajmlar farqi — bo'yi $20 - 12$ cm, eni 10 cm bo'lgan bo'lakning hajmi.)

14. **Qiyinroq.** A idish tubining yuzi 400 cm², undagi suv sathi 45 cm. B idish tubining yuzi 500 cm², u bo'sh. A dan B ga suv quyilib, ikkala idishdagi suv sathi tenglashdi. (Shunda suv go'yo tubi $400 + 500$ cm² bo'lgan bitta idishda turgandek bo'ladi.)
a) Suv sathi qanday balandlikda? b) B ga necha litr suv o'tdi?

15. **Amaliy ish.** Uyingizdagi biror qutining bo'yi, eni va balandligini o'lchang. Qutini uch xil qo'yib, har holatda asosining yuzini toping. Har holatda asosining yuzi × balandligi bir xil son chiqdimi?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $\frac{576}{12} = 48$ cm². b) $\frac{560}{5} = 112$ m². c) $\frac{264}{4} = 66$ cm².

2-masala

a) $\sqrt[3]{27000} = 30, 30 \cdot 30 = 900$ cm². b) $\sqrt[3]{9261} = 21, 21 \cdot 21 = 441$ cm². c) $\sqrt[3]{12167} = 23, 23 \cdot 23 = 529$ m².

3-masala

Old yoq bo'yi va balandligidan iborat, unga kirmagan qirra — eni: $\frac{36000}{40} = 900$ cm².

4-masala

A: $\frac{420}{12} = 35$ cm². B: $\frac{960}{40} = 24$ cm. C: $\frac{231}{21} = 11$ cm. D: $\frac{600}{15} = 40$ cm².

5-masala

a) $\frac{1125}{45} = 25$ cm². b) $\sqrt{25} = 5$ cm.

6-masala

Kvadrat yuzi $= \frac{1350}{6} = 225$ cm², tomoni $\sqrt{225} = 15$ cm.

7-masala

a) Quticha qirrasi $\sqrt[3]{125} = 5$ cm, katta qutiniki $2 \cdot 5 = 10$ cm, hajmi $10 \cdot 10 \cdot 10 = 1000$ cm³. b) $1000 : 125 = 8$ marta.

8-masala

a) Bitta kub: $864 : 4 = 216$ cm³, qirrasi $\sqrt[3]{216} = 6$ cm. b) Brusning bo'yi $4 \cdot 6 = 24$ cm.

9-masala

$\frac{27}{18} = 1{,}5$ m.

10-masala

a) 24 l $= 24\,000$ cm³, balandligi $\frac{24000}{800} = 30$ cm. b) 18 l $= 18\,000$ cm³, sath $\frac{18000}{800} = 22{,}5$ cm.

11-masala

To'la bak: $18 + 27 = 45$ l $= 45\,000$ cm³. Asosi $30 \cdot 30 = 900$ cm². Balandligi $\frac{45000}{900} = 50$ cm.

12-masala

Asosi $48 \cdot 25 = 1200$ cm². Suv $3 \cdot 6 = 18$ l $= 18\,000$ cm³. a) $\frac{18000}{1200} = 15$ cm. b) Bo'sh qism $32 - 15 = 17$ cm, $1200 \cdot 17 = 20\,400$ cm³ $= 20{,}4$ l.

13-masala

Tubi $2 \cdot 1{,}5 = 3$ m². Sath $\frac{1{,}2}{3} = 0{,}4$ m ga tushdi. Endi $0{,}9 - 0{,}4 = 0{,}5$ m.

14-masala

Sharbat sathi $20 : 2 = 10$ cm, hajmi $30 \cdot 20 \cdot 10 = 6000$ cm³. Ikkinchi idish tubi $50 \cdot 20 = 1000$ cm². Sath $\frac{6000}{1000} = 6$ cm.

15-masala

Tubi $60 \cdot 40 = 2400$ cm². a) $8 \cdot 9 = 72$ l $= 72\,000$ cm³, sath $\frac{72000}{2400} = 30$ cm. b) Idish $2400 \cdot 50 = 120\,000$ cm³ $= 120$ l, $120 : 8 = 15$ minut.

16-masala

Kub qirrasi $\sqrt[3]{27} = 3$ cm, yog'i $3 \cdot 3 = 9$ cm². 3 ta kubda $3 \cdot 6 = 18$ ta yoq, 2 ta joyda $2 \cdot 2 = 4$ ta yoq ichkarida qoladi. Tashqarida $18 - 4 = 14$ ta yoq: $14 \cdot 9 = 126$ cm².

17-masala

To'la yashik: $2 \cdot 10 = 20$ l $= 20\,000$ cm³. Kvadrat yuzi $\dfrac{20000}{50} = 400$ cm², eni $\sqrt{400} = 20$ cm.

18-masala

Suv: $900 \cdot 32 = 28\,800$ cm³. a) $\dfrac{28800}{600} = 48$ cm. b) Bo'sh qism $40 - 32 = 8$ cm, $900 \cdot 8 = 7200$ cm³ $= 7{,}2$ l.

Mustaqil ish

1-masala

a) $\dfrac{468}{4} = 117$ cm². b) $\dfrac{448}{16} = 28$ m².

2-masala

$\sqrt[3]{10648} = 22$ cm, yog'i $22 \cdot 22 = 484$ cm².

3-masala

Asosi $\dfrac{1620}{5} = 324$ cm², tomoni $\sqrt{324} = 18$ cm.

4-masala

Plita hajmi $30 \cdot 20 \cdot 6 = 3600$ cm³. Kvadrat yuzi $\dfrac{3600}{36} = 100$ cm², tomoni $\sqrt{100} = 10$ cm.

5-masala

Katta kub qirrasi $\sqrt[3]{13824} = 24$ cm. Kichik kubniki $24 : 4 = 6$ cm, hajmi $6 \cdot 6 \cdot 6 = 216$ cm³.

6-masala

Tubi $70 \cdot 40 = 2800$ cm². a) Bo'sh qism balandligi $\dfrac{42000}{2800} = 15$ cm, suv sathi $45 - 15 = 30$ cm. b) Suv $2800 \cdot 30 = 84\,000$ cm³ $= 84$ l.

7-masala

Tubi $40 \cdot 30 = 1200$ cm². Sath $\frac{6000}{1200} = 5$ cm ga ko'tarildi: $12 + 5 = 17$ cm.

8-masala

Idishdagi sut sathi $\frac{2}{3} \cdot 30 = 20$ cm, sut $50 \cdot 40 \cdot 20 = 40\,000$ cm³. Tog'ora tubi $50 \cdot 50 = 2500$ cm². Sath $\frac{40000}{2500} = 16$ cm.

9-masala

Tubi $80 \cdot 45 = 3600$ cm². a) Suv $3600 \cdot 40 = 144\,000$ cm³ $= 144$ l, $144 : 9 = 16$ minut. b) 4 minutda $9 \cdot 4 = 36$ l $= 36\,000$ cm³ oqdi, sath $\frac{36000}{3600} = 10$ cm ga tushdi: $40 - 10 = 30$ cm.

10-masala

$40 : 10 = 4$, $30 : 6 = 5$, $20 : 5 = 4$. $4 \cdot 5 \cdot 4 = 80$ ta quticha.

11-masala

Kub qirrasi $\sqrt[3]{216} = 6$ cm. Hosil bo'lgan kuboid: bo'yi $10 + 6 = 16$ cm, eni 6 cm, balandligi 6 cm. Hajmi $16 \cdot 6 \cdot 6 = 576$ cm³.

12-masala

a) Kubning 6 ta yog'i bor, lekin hajm 6 ta yoqning yig'indisi emas. To'g'risi: qirrasi $\sqrt[3]{1000} = 10$ cm, yog'i $10 \cdot 10 = 100$ cm². b) Kubning qirrasi uchun kub ildiz $\sqrt[3]{1000} = 10$ kerak, kvadrat ildiz emas.

13-masala

Farq bo'lagining asosi: $(20 - 12) \cdot 10 = 80$ cm². Balandligi $\frac{640}{80} = 8$ cm.

14-masala

Suv $400 \cdot 45 = 18\,000$ cm³. Ikkala tub birga $400 + 500 = 900$ cm². a) $\frac{18000}{900} = 20$ cm. b) B ga $500 \cdot 20 = 10\,000$ cm³ $= 10$ l o'tdi.

15-masala

Javob har kimda har xil. Uchala holatda ham asosining yuzi × balandligi qutining hajmiga teng, ya'ni bir xil son chiqadi.